SMRA 2026

Clinically Calibrated Large Language Model Reasoning for Zero-Shot Cardiac MRI Pathology Classification from Segmentation-Derived Metrics

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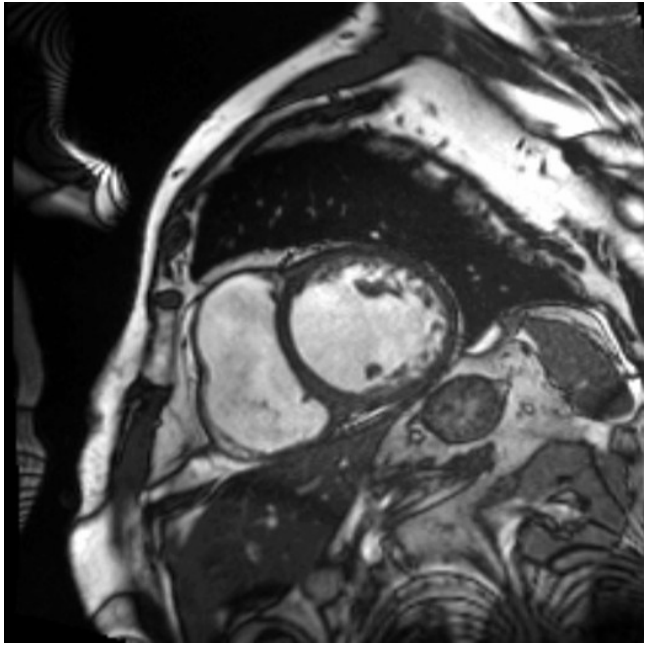
1. Weldon School of Biomedical Engineering, Purdue University, West Lafayette, USA

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Can an LLM identify cardiac disease from three numbers?

14%	352	304
EF	EDV (mL)	ESV (mL)

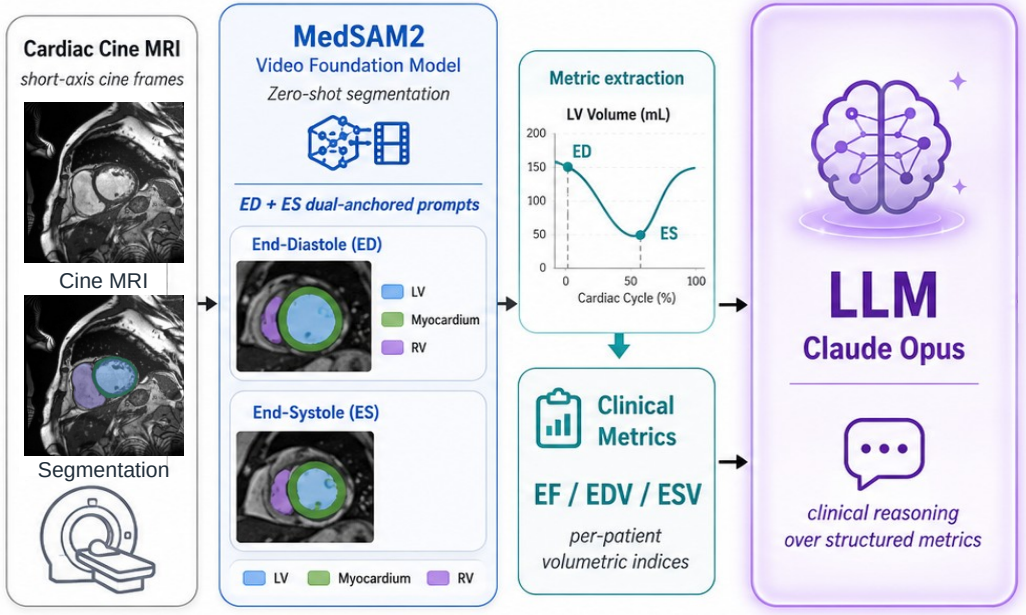
Reference LV measurements



ACDC patient017

What did we ask the LLM to do?

Numerical MRI measurements and clinical instructions. No task-specific fine-tuning.



Prompt	What we ask
Direct	Choose one disease label
Stepwise	Assess EF, then volume, using supplied thresholds
Calibrated	Use revised thresholds and one worked DCM example

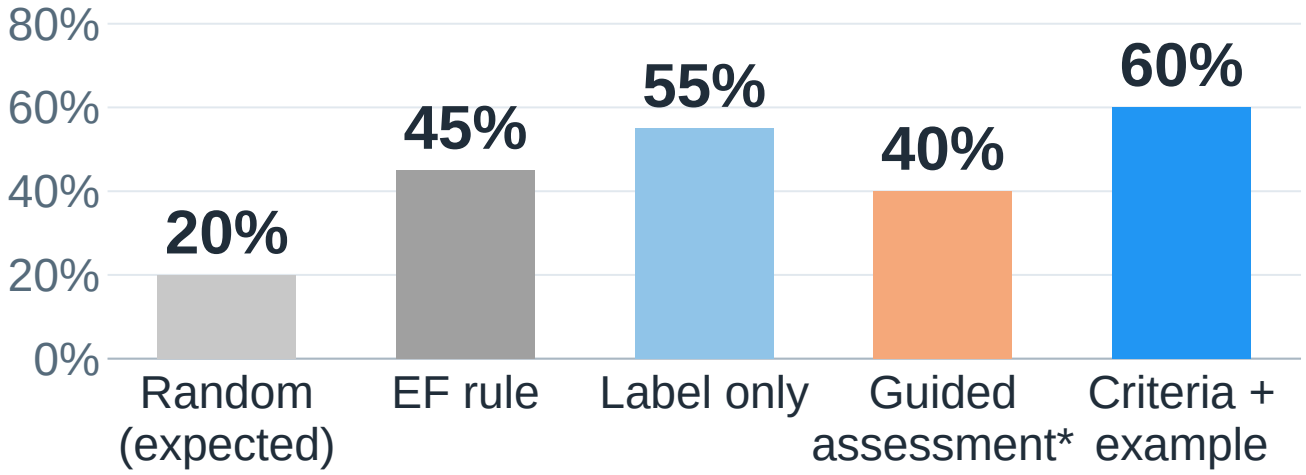
Prompt example: EF 28%, EDV 310 mL, ESV 223 mL
Reduced function + dilated cavity. Label: DCM.

LLM input: EF, EDV and ESV as text

Reference LV metrics. 20 ACDC patients across 5 classes, with 4 cases per class.

Revised criteria + an example reached 60%

Same LLM and reference measurements, different text instructions



12/20

correct cases

0.510

macro-F1

Exploratory results: one more correct case than the label-only prompt.

* 8/20 stepwise outputs lacked a final label, so parsing affects this comparison.

Where the LLM succeeded, and what it missed

Confusion matrix for calibrated prompting, 20 ACDC cases

Predicted class

True class	NOR	DCM	HCM	MINF	RV
NOR	4	0	0	0	0
DCM	0	4	0	0	0
HCM	1	0	0	2	1
MINF	0	1	0	3	0
RV	2	0	0	1	1

8 / 8

Normal and DCM cases correctly labeled

0 / 4

HCM cases correctly labeled

Inputs omit wall thickness and right-ventricular measurements.

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RESEARCH CONTRIBUTION

Clinical guidance through language

An LLM interpretation layer for quantitative MRI, without task-specific fine-tuning.

Next: richer biomarkers and independent validation.

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github.com/zhuoanli/llm-cardiac-pathology

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Welcome to connect and discuss interesting research !

Project page